Band-1 Hadamard Order 4948 — Miyamoto (Paley-II seed H_1236); in SPK via Paley-II (q=2473); Gap 2052 Still OPEN

H_1236/C_1238/H_4948 PASS; optional Paley-II H_4948 PASS; 4948 in SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4916_Miyamoto`; `2026-09-15_Band1_Hadamard_4804_Miyamoto` / `4036` (in-SPK Paley-II pattern).
Lab. `/workspace/experiments/band1/hadamard_4948_miyamoto/`.
Cutoff. 2026-09-15T18:27:29Z (~2:27PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1236 and 4948 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1236} ($q = 617$), and optionally via classical Paley-II ($q = 2473$). Paley-I direct for H_{1236} is **SKIP** ($q = 1235$ composite). Double Paley-I half = 618 is **FAIL** ($q = 617 \equiv 1 \pmod 4$; Paley-I needs $\equiv 3$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4948 is **in SPK via Paley-II** (prime $q = 2473$): Paley-I FAIL ($q = 4947$ composite), Paley-II PASS ($q = 2473$ prime $\equiv 1 \pmod 4$), Kronecker FAIL (4×1237). Same honesty pattern as 4804/4036 (in SPK, not Miyamoto-only). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4916 on box. This increment ships **order 4948** ($4948 = 4 \cdot 1237$, 1237 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Paley-II seed** H_{1236} ($q = 617$). Symmetric Paley conference C_{1238} over prime 1237; $H_{4948} = \text{Miyamoto}(H_{1236}, C_{1238})$. Classical Paley-II path for order 4948 ($q = 2473$) also verified on box. Unit tests **57/57 PASS**. SPK next gap remains **2052** (OPEN). Order 4948 is **in SPK via Paley-II** (NOT Miyamoto-only). Verified H_{1236} enters SPK as a Paley-II seed. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4948; Miyamoto; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; in-SPK via Paley-II; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1236} constructed and verified (Paley type II, $q = 617$). Paley-I direct **SKIP** ($q = 1235$ composite). Double Paley-I **FAIL** ($q = 617 \equiv 1 \pmod 4$). 2. C_{1238} constructed and verified (Paley conference, $q = 1237$): $CC^\top = 1237I$, zero diagonal. 3. H_{4948} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. Optional: H_{4948} also constructed via Paley-II ($q = 2473$) and verified. 5. SPK: $4948 \in$ SPK via Paley-II (prime $q = 2473$). Paley-I / Kronecker paths for 4948 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{1236} enters SPK as Paley-II seed. Prior 1228 (Paley-II) and 4804 (Paley-II 7^4) remain in SPK. Prior 4916/4868 remain Miyamoto-only (excluded from SPK seeds).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4948 = 4 \cdot 1237$
$q = 1237$	prime, $1237 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1236}
Needed conference	order $1238 = q + 1$
H_{1236} path (used)	Paley-II: $2(q+1) = 1236 \Rightarrow q = 617$ prime $\equiv 1 \pmod 4$
H_{1236} Paley-I direct	**SKIP**: $q = 1235$ composite
H_{1236} Double Paley-I	**FAIL**: half $= 618$, $q = 617 \equiv 1 \pmod 4$ (needs $\equiv 3$)
Conference path	Symmetric Paley over prime 1237
Paley-I for 4948	**REJECTED**: $q' = 4947$ composite
Paley-II for 4948	**PASS**: $q' = 2473$ prime $\equiv 1 \pmod 4 \rightarrow$ $n = 2(q+1) = 4948$ **in SPK**

Quantity	Value
Kronecker for 4948	**REJECTED**: only 4×1237; $1237 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1236}	Paley type II, $q = 617$	0	1.1103	**PASS**
H_{1236} Paley-I direct	$q = 1235$ composite	—	—	**SKIP**
H_{1236} Double Paley-I	half $= 618$, $q = 617 \equiv 1$	—	—	**FAIL** (needs $\equiv 3$)
C_{1238}	Paley conference, $q = 1237$	0	1.5832	**PASS**
H_{4948}	Miyamoto(H_{1236}, C_{1238})	0	54.4394	**PASS**
H_{4948} (opt)	Paley-II, $q = 2473$	0	56.8337	**PASS**

Primary pipeline wall (incl. unit tests + optional Paley-II + SPK refresh) ≈ 117.3957 s. Compressed ± 1 matrices saved as `H1236_pm1.npz`, `C1238_pm1.npz`, `H4948_pm1.npz`, `H4948_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 4916/4868/4804/4036/3908/… + **1020/1116/1228** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128/600/1200/608/12** Double Paley-I + **4804** Paley-II prime power 7^4; **exclude** Miyamoto-only 4916/4868/4612/4516/4468/4388/4372/4 $\cup$ verified H_{1236} (Paley-II) $\cup$ verified H_{4948} (Paley-II classical) $\cup$ Sylvester $\cup$ Paley I/II (incl. prime powers), closed under Kronecker (limit 5000). Order 4948 is **in SPK via Paley-II** (folded into SPK extras).

Quantity	Value
Known count	1047
4948 in SPK	**True** (via Paley-II q=2473)
1236 in SPK	**True** (Paley-II seed)
1228 in SPK	**True** (prior Paley-II)

Quantity	Value
4804 in SPK	**True** (Paley-II prime power 7^4)
4916 in SPK	**False** (prior Miyamoto-only)
4868 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 4948 is **in SPK via Paley-II** (NOT a Miyamoto-only bonus).
- Gap 2052 is **not** filled by this ship.

7. Reproduce

```
cd /workspace/experiments/band1/hadamard_4948_miyamoto
python3 verify_hadamard_4948.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H1236_pm1.npz`, `C1238_pm1.npz`, `H4948_pm1.npz`, `H4948_paleyII_pm1.npz`, this paper.

8. Next optional

After $q = 1237$: **q=1249→n=4996** Kronecker(4×312), or **q=1289→n=5156** Double(PaleyI half=644), or method-attack SPK gap **2052**, or Track B privacy / Object 1 meter await.